

GAVE Challenge Team Technical Report

Team Name: R2-V2

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Abstract

This document presents the technical report for the R2-V2 team participating in the Generalized Analysis of Vessels in Eye (GAVE) Challenge. The report details our approach, methodologies, and experimental configurations employed in addressing the challenge Tasks 1 and 2, which involve the segmentation of blood vessels and their classification into arteries and veins in retinal images. No work was done for Task 3, so it is not included in this report.

1 Proposed Method

1.1 Model Framework

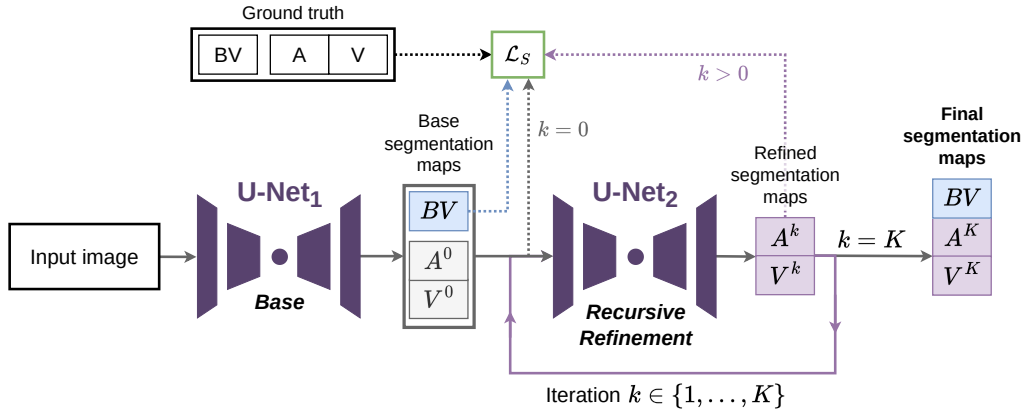


Figure 1: Proposed method, based on [1].

1.2 Structure Detail Description

Our approach is based on the RRWNet architecture [1], which is depicted in Figure 1. The architecture consists of two main components: the Base segmentation module and the Recursive Refinement (RR) module. Both components are based on a standard U-Net architecture [2], shown in Figure 2, only adapting the number of input and output channels.

The Base segmentation module generates an initial segmentation of blood vessels (BV), arteries, and veins from the input image. Then, the RR module refines the initial arteries and veins (A/V) segmentation in a recursive and iterative manner, improving the segmentation quality. The final output of the model is the combination of the initial BV segmentation from the Base segmentation module and the refined A/V segmentation from the RR module.

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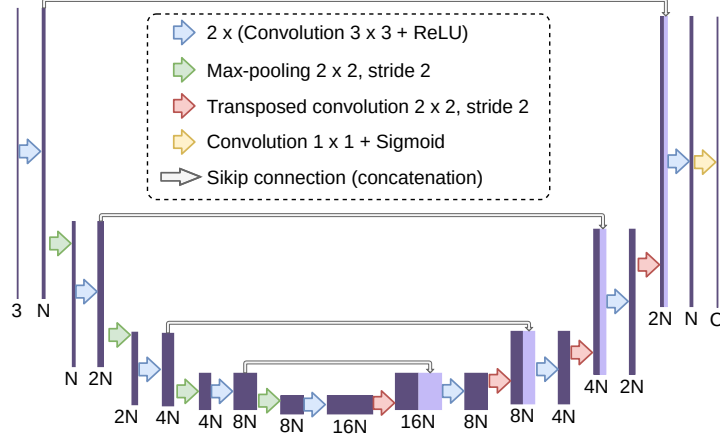


Figure 2: U-Net architecture [2].

1.3 Input/Output (Data Flow)

Let $\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$ be the input RGB color fundus image, where H and W are its height and width, respectively. To obtain the final prediction $\hat{\mathbf{y}} \in \mathbb{R}^{3 \times H \times W}$, we apply the network $f(\mathbf{x}, \theta, K)$ to the input image $\mathbf{x}$, where θ represents the learnable parameters of the network and K denotes the number of iterations performed by the RR subnetwork. Thus, the output of the network $\hat{\mathbf{y}}_k$ at an arbitrary iteration k can be defined as

$$\begin{aligned} \hat{\mathbf{y}}_k &= f(\mathbf{x}, \theta, k) \\ &= \begin{cases} f_R(f(\mathbf{x}, \theta, k-1), \theta^R)_{a,v} \oplus f_B(\mathbf{x}, \theta^B)_{bv}, & k > 0 \\ f_B(\mathbf{x}, \theta^B), & k = 0 \end{cases} \end{aligned} \quad (1)$$

where f_B and f_R represent the Base and RR subnetworks with parameters θ^B and θ^R , respectively; the subscripts a, v, and bv denote the channels corresponding to the segmentation maps of the different structures; and $\oplus$ denotes the concatenation operation in the channel dimension.

2 Experiment Configuration

2.1 Datasets

For training the model, we used the training set provided by the GAVE Challenge along with the following public datasets: RITE [3], [4], HRF [5], [6], and LES-AV [7], Fundus-AVSeg [8], AFIO [9], and DualModal2019 [10]. All these datasets provide annotations for arteries and veins as well as for the regions of interest (ROIs) (i.e., the retinal area visible in the image). The combination of these datasets provides a diverse set of 386 color fundus images.

2.2 Datasets Split Method

For the development of the model, we randomly split the combined dataset into training, validation, and test sets (70%, 15%, and 15% of the total data, respectively). However, for the preliminary submission, we trained the model using all the available data. In this way, the final models were trained using the 386 images.

2.3 Data Augmentation

To enhance the diversity of the training data and improve the generalization of the model, we applied strong data augmentation. The full list of augmentations is detailed below. In each case, the distribution from which the augmentation parameters are sampled (if applicable) is indicated, along with the probability p of applying the augmentation operation.

- Horizontal and vertical flips ($p = 0.5$ each).

- Random rotations with angle $\theta \sim \mathcal{U}(-180^\circ, 180^\circ)$ ($p = 1$).
- Random scaling with factor $\alpha \sim \mathcal{U}(0.7, 1.4)$ ($p = 1$).
- Random shearing with angle $\gamma \sim \mathcal{U}(-25^\circ, 25^\circ)$ ($p = 1$).
- Random hue, saturation, and value (HSV) color jittering ($p = 1$):
 - $\Delta H \sim \mathcal{U}(-0.02, 0.02)$.
 - $\Delta S \sim \mathcal{U}(-0.2, 0.2)$.
 - $\Delta V \sim \mathcal{U}(-0.2, 0.2)$.
 - For H and V, there is also a multiplicative factor $m \sim \mathcal{U}(0.8, 1.2)$.
- Random cutout ($p = 0.8$):
 - 16 patches of size 0.04×0.04 of the image size.
 - Top left coordinates (i, j) sampled uniformly across the image, so that $i \sim \mathcal{U}(0, W)$ and $j \sim \mathcal{U}(0, H)$, where W and H are the width and height of the image, respectively.
 - The value inside each patch is sampled from $\mathcal{U}(0.4, 0.6)$.

For inference, we use test-time augmentation (TTA) with horizontal and vertical flips as well as 90-degree rotations, resulting in a total of 8 different augmentations (including the original image).

2.4 Preprocessing

As preprocessing, we applied channel-wise global contrast enhancement and local intensity normalization (CGCELIN) (as proposed in [11]) followed by contrast limited adaptive histogram equalization (CLAHE) [12]. In particular, CGCELIN is defined as follows:

$$I_{\text{norm}}^C = \sigma_0 \frac{I^C - I_l^C}{\sigma_{I^C - I_l^C}},$$

where I_{norm}^C is the normalized channel C , I^C is the channel C of the original input image, I_l^C is the low-pass filtered channel C and $\sigma_{I^C - I_l^C}$ is the global standard deviation of the channel resulting from the subtraction of the low-pass filtered channel I_l^C to the original channel I^C .

As in [11], we use $\sigma_0 = 1$, with a Gaussian filter with zero mean and standard deviation $\sigma = 10$. For training, the input images were cropped to the ROI using the provided masks and resized to a fixed width of 1408 pixels, maintaining the original aspect ratio. For inference in the GAVE validation set, no resizing nor cropping was applied, and the images were processed at their original resolution.

2.5 Training Strategy

Loss Function

To train the network, we build on the loss function proposed in [1]. Specifically, the total loss $\mathcal{L}$ combines the segmentation errors from each iteration $k \in \{0, \dots, K\}$ with different weights. This loss is defined as:

$$\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) = \sum_{k=0}^K w_k \mathcal{L}_S(\hat{\mathbf{y}}_k, \mathbf{y}), \quad (2)$$

where w_k is the weight of the loss at iteration k , $\hat{\mathbf{y}}_k$ is the output of the network at iteration k (see Eq. 1), $\mathbf{y}$ is the ground truth (GT), and $\mathcal{L}_S$ is the base segmentation loss function. The weighting scheme is the same as in [1], giving a higher weight to the error of the first iteration ($k = 0$), which solely relies on the Base subnetwork. Specifically, the weight w_k at iteration k is defined as

$$w_k = \begin{cases} 1, & k = 0 \\ \frac{1}{Z} \sum_{k=1}^K k \mathcal{L}_S(\hat{\mathbf{y}}_k, \mathbf{y}), & k > 0 \end{cases} \quad (3)$$

with the normalization factor $Z = \sum_{k=1}^K k = \frac{K(K+1)}{2}$.

While the general formulation is the same as in [1], the specific definition of $\mathcal{L}_S$ is different, as explained next. In particular, we adapt the loss function to focus on multiple aspects of the segmentation task, including vessel presence, A/V discrimination, A/V mutual exclusion, crossing identification, and background consistency. Formally, the total base loss function $\mathcal{L}_S$ can be defined as:

$$\mathcal{L}_S = \lambda_{bv} \cdot \mathcal{L}_{bv} + \lambda_{av} \cdot \mathcal{L}_{av} + \lambda_{mx} \cdot \mathcal{L}_{mx} + \lambda_{cr} \cdot \mathcal{L}_{cr} + \lambda_{bg} \cdot \mathcal{L}_{bg}, \quad (4)$$

where each $\mathcal{L}_{(\cdot)}$ component corresponds to a specific aspect of the segmentation task, and $\lambda_{(\cdot)}$ are the corresponding weights.

The definition of each component is shown below. However, we will first introduce some notation to facilitate their explanation. Since the components operate on specific structure predictions and regions of the image, we use subscripts in the predicted and GT variables ($\hat{\mathbf{y}}$ and $\mathbf{y}$, respectively) to indicate the selected channel: a: artery, v: vein, bv: blood vessel (artery + vein). Similarly, the superscripts are used to indicate the selected region (pixels) of the image: a: arteries, v: veins, bv: blood vessels (artery + vein), bg: background, cr: crossings, roi: region of interest, and a $\vee$ v: arteries or veins (i.e., bv excluding crossings). Moreover, $\odot$ denotes the element-wise multiplication, $\mathbf{0}$ and $\mathbf{1}$ are matrices of zeros and ones with the same size as the corresponding prediction, respectively, and $\text{avg}(\cdot)$ denotes the average value of all elements in the matrix. With this notation, the different components of the loss function are defined as follows:

- **Vessel Presence Loss:**

$$\mathcal{L}_{bv} = \text{BCE}(\hat{\mathbf{y}}_{bv}^{\text{roi}}, \mathbf{y}_{bv}^{\text{roi}}) \quad (5)$$

- **Artery/Vein Discrimination Loss:**

$$\mathcal{L}_{av} = \text{BCE}(\hat{\mathbf{y}}_a^{\text{bv}}, \mathbf{y}_a^{\text{bv}}) + \text{BCE}(\hat{\mathbf{y}}_v^{\text{bv}}, \mathbf{y}_v^{\text{bv}}) \quad (6)$$

- **Mutual Exclusion Loss:**

$$\mathcal{L}_{mx} = \text{avg}(\hat{\mathbf{y}}_a^{\text{a}\vee\text{v}} \odot \hat{\mathbf{y}}_v^{\text{a}\vee\text{v}}) \quad (7)$$

- **Crossings Handling Loss:**

$$\mathcal{L}_{cr} = \text{BCE}\left(\frac{\hat{\mathbf{y}}_a^{\text{cr}} + \hat{\mathbf{y}}_v^{\text{cr}}}{2}, \mathbf{1}\right) \quad (8)$$

- **Background Consistency Loss:**

$$\mathcal{L}_{bg} = \text{BCE}(\hat{\mathbf{y}}_a^{\text{bg}}, \mathbf{0}) + \text{BCE}(\hat{\mathbf{y}}_v^{\text{bg}}, \mathbf{0}) \quad (9)$$

Optimization Details

The models were trained using the Adam [13] optimizer with a learning rate of $1e-4$ and decay rates $\beta_1 = 0.9$ and $\beta_2 = 0.999$, as in [13]. The batch size used is 1, to allow for variable input sizes. The final models were trained for 200 000 iterations (≈ 518 epochs).

2.6 Model Selection and Postprocessing

For the preliminary submission, we trained two separate models: f_{BV} , which focuses on segmenting BV and takes as input both the preprocessed and the original images (6 channels), and f_{AV} , which focuses on classifying arteries and veins and takes as input only the preprocessed image (3 channels). Both models use $K = 6$ iterations in the RR module, and are trained using the same loss function. However, they differ in the specific weights used for each component of the loss function, in order to specialize each model for its specific task.

For training f_{BV} , the idea was to adjust the weights to put more emphasis on the discrimination between background and vessels. For this reason, the weights of λ_{bv} , λ_{av} , and λ_{bg} were set proportionally to the number of pixels belonging to the ROI of each loss component scaled by some factors. In particular, we use factors of 1.0, 2.0, and 0.5 for λ_{bv} , λ_{av} , and λ_{bg} , respectively. Differently, the weights for the other two components, λ_{mx} and λ_{cr} , are fixed to 0.5 and 0.2, respectively. Formally, the weights for this model can be defined as follows:

$$\lambda_{bv} = \frac{N_{bv}}{N_{\text{roi}}}, \quad \lambda_{av} = 2 \cdot \frac{N_{\text{a}\vee\text{v}}}{N_{\text{roi}}}, \quad \lambda_{bg} = 0.5 \cdot \frac{N_{bg}}{N_{\text{roi}}}, \quad \lambda_{mx} = 0.5, \quad \lambda_{cr} = 0.2, \quad (10)$$

where N denotes the number of pixels in the corresponding region, which is indicated by the subscript. As can be seen, the higher weight is given to the background consistency loss (since around 80-90% of the pixels belong to the background), followed the vessel presence loss.

On the other hand, for training f_{AV} , we set empirically the weights to the following values, in order to focus more on the A/V discrimination:

$$\lambda_{\text{vessel}} = 1.0, \quad \lambda_{\text{av}} = 2.0, \quad \lambda_{\text{bg}} = 0.5, \quad \lambda_{\text{mx}} = 0.5, \quad \lambda_{\text{cr}} = 0.5 \quad (11)$$

In this way, the highest weight is given to the A/V discrimination loss.

The final segmentation was obtained by combining the BV segmentation from f_{BV} and the A/V segmentation from f_{AV} . However, before combining them, we apply a postprocessing step to improve the sensitivity of the BV segmentation. In particular, we set the segmented BV to include all the pixels classified as BV, arteries, or veins. This is done by applying the following operation:

$$\hat{\mathbf{y}}_{\text{bv}} = \max(\hat{\mathbf{y}}_{\text{bv}}, \text{clip}(\hat{\mathbf{y}}_{\text{a}} + \hat{\mathbf{y}}_{\text{v}}, 0, 1)) , \quad (12)$$

where $\hat{\mathbf{y}}_{\text{bv}}$, $\hat{\mathbf{y}}_{\text{a}}$, and $\hat{\mathbf{y}}_{\text{v}}$ are the predicted BV, artery, and vein maps, respectively, $\max(\cdot, \cdot)$ is the element-wise maximum operation, and $\text{clip}(\cdot, 0, 1)$ limits the values to the range $[0, 1]$. No postprocessing is applied to the output segmentation maps of arteries and veins coming from the A/V model, except for setting to zero the pixels outside the ROI using the provided masks, which is done for all the maps.

Consequently, the final output of the model is given by the concatenation of the BV map from the BV model and the artery and vein maps from the A/V model:

$$\hat{\mathbf{y}} = (\hat{\mathbf{y}}_{\text{bv}} \oplus \hat{\mathbf{y}}_{\text{a,v}}) \odot \mathbf{m}_{\text{roi}} , \quad (13)$$

where $\hat{\mathbf{y}}_{\text{bv}}$ is the postprocessed BV map from f_{BV} , $\hat{\mathbf{y}}_{\text{a,v}}$ are the artery and vein maps from f_{AV} , and $\mathbf{m}_{\text{roi}}$ is the binary mask of the ROI.

2.7 Technical Stack

The implementation was done using Python 3.12, PyTorch 2.2¹, torchvision 0.17², CUDA 12.1³, NumPy 1.26⁴, scikit-image 0.22⁵, and scipy 1.12⁶. The training was performed using a NVIDIA A100 GPU with 80 GB of memory.

3 GitHub

The inference code for the proposed method is publicly available at https://github.com/j-morano/R2-V2_materials.

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²<https://pytorch.org/vision/> [Accessed: Aug. 25, 2025]

³<https://developer.nvidia.com/cuda-toolkit> [Accessed: Aug. 25, 2025]

⁴<https://numpy.org/> [Accessed: Aug. 25, 2025]

⁵<https://scikit-image.org/> [Accessed: Aug. 25, 2025]

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